Attendance Requirement Statement for CS 156a Discussion Sessions

Please complete & attach as the 1st page of your submitted final set

[required for the final set to be graded and credited]

Name: _____

I have attended at least 8 of the 18 discussion sessions in class (at least 7 if I submitted the bonus exercise).

Signature: _____

Fall 2025

CS/CNS/EE 156a Learning Systems
Caltech - Fall 2025
https://caltech.instructure.com/courses/8882
(Learning From Data campus version)

Final Set

Due **Friday**, December 5, 2025, at 2:00 PM PST

*Definitions and notation follow the lectures. All questions have multiple-choice answers (***[a], [b], [c], ...***). Collaboration is not allowed in this final set. You can ask questions on the course forum for clarification only.* **You cannot discuss the problems with anyone**. *The open book/chatGPT policies in the initial handout apply to the final set as well.*

Note about the final set

- Be sure to give yourself enough time to complete this final set. There are twice as many problems in this set as there are in a homework set, and some problems require packages that will need time to get to work properly. A bit more time is given for this set compared to the homework (due date is Friday of next week).
- Problems cover different parts of the course. You are encouraged to consult your previous course work. To facilitate your search for relevant parts, an indexed version of the lecture video segments can be found at the Machine Learning Video Library:

  http://work.caltech.edu/library

- Answer each question by deriving the answer (carries 6 points) then selecting from the multiple-choice answers (carries 4 points). You can select 1 or 2 of the multiple-choice answers for each question, but you will get 4 or 2 points, respectively, for a correct answer. See the initial "**Course Description and Policies**" handout for important details.
- This is a no-collaboration set. You can only ask questions for clarification on the discussion forum, but otherwise you should not discuss the problems at all with anyone before the final set is due.

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- **Nonlinear transforms**

1. The polynomial transform of order $Q = 10$ applied to $\mathcal{X}$ of dimension $d = 2$ results in a $\mathcal{Z}$ space of what dimensionality (not counting the constant coordinate $x_0 = 1$ or $z_0 = 1$)?

[**a**] 12

[**b**] 20

[**c**] 35

[**d**] 100

[**e**] None of the above

- **Bias and Variance**

2. Recall that the average hypothesis $\bar{g}$ was based on training the same model $\mathcal{H}$ on different data sets $\mathcal{D}$ to get $g^{(\mathcal{D})} \in \mathcal{H}$, and taking the expected value of $g^{(\mathcal{D})}$ w.r.t. $\mathcal{D}$ to get $\bar{g}$. Which of the following models $\mathcal{H}$ could result in $\bar{g} \notin \mathcal{H}$?

[**a**] A singleton $\mathcal{H}$ ($\mathcal{H}$ has one hypothesis)

[**b**] $\mathcal{H}$ is the set of all constant, real-valued hypotheses

[**c**] $\mathcal{H}$ is the linear regression model

[**d**] $\mathcal{H}$ is the logistic regression model

[**e**] None of the above

- **Overfitting**

3. Which of the following statements is *false*?

[**a**] If there is overfitting, there must be two or more hypotheses that have different values of E_{in}.

[**b**] If there is overfitting, there must be two or more hypotheses that have different values of E_{out}.

[**c**] If there is overfitting, there must be two or more hypotheses that have different values of $(E_{\text{out}} - E_{\text{in}})$.

[**d**] We can always determine if there is overfitting by comparing the values of $(E_{\text{out}} - E_{\text{in}})$.

[**e**] We cannot determine overfitting based on one hypothesis only.

4. Which of the following statements is true?

[**a**] Deterministic noise cannot occur with stochastic noise.

[**b**] Deterministic noise does not depend on the hypothesis set.

[**c**] Deterministic noise does not depend on the target function.

[**d**] Stochastic noise does not depend on the hypothesis set.

[**e**] Stochastic noise does not depend on the target distribution.

• Regularization

5. The regularized weight $\mathbf{w}_{\text{reg}}$ is a solution to:

$$\text{minimize} \quad \frac{1}{N}\sum_{n=1}^{N}(\mathbf{w}^{\mathrm{T}}\mathbf{x}_n - y_n)^2 \quad \text{subject to} \quad \mathbf{w}^{\mathrm{T}}\Gamma^{\mathrm{T}}\Gamma\mathbf{w} \leq C,$$

where Γ is a Tikhonov matrix. If $\mathbf{w}_{\text{lin}}^{\mathrm{T}}\Gamma^{\mathrm{T}}\Gamma\mathbf{w}_{\text{lin}} \leq C$, where $\mathbf{w}_{\text{lin}}$ is the linear regression solution, then what is $\mathbf{w}_{\text{reg}}$?

[**a**] $\mathbf{w}_{\text{reg}} = \mathbf{w}_{\text{lin}}$

[**b**] $\mathbf{w}_{\text{reg}} = \Gamma\mathbf{w}_{\text{lin}}$

[**c**] $\mathbf{w}_{\text{reg}} = \Gamma^{\mathrm{T}}\Gamma\mathbf{w}_{\text{lin}}$

[**d**] $\mathbf{w}_{\text{reg}} = C\Gamma\mathbf{w}_{\text{lin}}$

[**e**] $\mathbf{w}_{\text{reg}} = C\mathbf{w}_{\text{lin}}$

6. Soft-order constraints that regularize polynomial models can be

[**a**] written as hard-order constraints

[**b**] translated into augmented error

[**c**] determined from the value of the VC dimension

[**d**] used to decrease both E_{in} and E_{out}

[**e**] None of the above is true

• Regularized Linear Regression

We are going to experiment with linear regression for classification on the processed US Postal Service Zip Code data set from Homework 8. Download the data (extracted features of intensity and symmetry) for training and testing:

http://www.amlbook.com/data/zip/features.train

http://www.amlbook.com/data/zip/features.test

(the format of each row is: **digit intensity symmetry**). We will train two types of binary classifiers; one-versus-one (one digit is class $+1$ and another digit is class -1, with the rest of the digits disregarded), and one-versus-all (one digit is class $+1$ and the rest of the digits are class -1). When evaluating E_{in} and E_{out}, use binary classification error. Implement the regularized least-squares linear regression for classification that minimizes

$$\frac{1}{N}\sum_{n=1}^{N}\left(\mathbf{w}^{\mathrm{T}}\mathbf{z}_n - y_n\right)^2 \;+\; \frac{\lambda}{N}\mathbf{w}^{\mathrm{T}}\mathbf{w}$$

where $\mathbf{w}$ includes w_0.

7. Set $\lambda = 1$ and do not apply a feature transform (i.e., use $\mathbf{z} = \mathbf{x} = (1, x_1, x_2)$). Which among the following classifiers has the lowest E_{in}?

[**a**] 5 versus all

[**b**] 6 versus all

[**c**] 7 versus all

[**d**] 8 versus all

[**e**] 9 versus all

8. Now, apply a feature transform $\mathbf{z} = (1, x_1, x_2, x_1x_2, x_1^2, x_2^2)$, and set $\lambda = 1$. Which among the following classifiers has the lowest E_{out}?

[**a**] 0 versus all

[**b**] 1 versus all

[**c**] 2 versus all

[**d**] 3 versus all

[**e**] 4 versus all

9. If we compare using the transform versus not using it, and apply that to '0 versus all' through '9 versus all', which of the following statements is correct for $\lambda = 1$?

[**a**] Overfitting always occurs when we use the transform.

[**b**] The transform always improves the out-of-sample performance by at least 5% (E_{out} with transform $\leq 0.95E_{\text{out}}$ without transform).

[**c**] The transform does not make any difference in the out-of-sample performance.

[d] The transform always worsens the out-of-sample performance by at least 5%.

[e] The transform improves the out-of-sample performance of '5 versus all,' but by less than 5%.

10. Train the '1 versus 5' classifier with $\mathbf{z} = (1, x_1, x_2, x_1x_2, x_1^2, x_2^2)$ with $\lambda = 0.01$ and $\lambda = 1$. Which of the following statements is correct?

[a] Overfitting occurs (from $\lambda = 1$ to $\lambda = 0.01$).

[b] The two classifiers have the same E_{in}.

[c] The two classifiers have the same E_{out}.

[d] When λ goes up, both E_{in} and E_{out} go up.

[e] When λ goes up, both E_{in} and E_{out} go down.

• Support Vector Machines

11. Consider the following training set generated from a target function $f : \mathcal{X} \to \{-1, +1\}$ where $\mathcal{X} = \mathbb{R}^2$

$$\begin{array}{ccc} \mathbf{x}_1 = (1,0), y_1 = -1 & \mathbf{x}_2 = (0,1), y_2 = -1 & \mathbf{x}_3 = (0,-1), y_3 = -1 \\ \mathbf{x}_4 = (-1,0), y_4 = +1 & \mathbf{x}_5 = (0,2), y_5 = +1 & \mathbf{x}_6 = (0,-2), y_6 = +1 \\ & \mathbf{x}_7 = (-2,0), y_7 = +1 & \end{array}$$

Transform this training set into another two-dimensional space $\mathcal{Z}$

$$z_1 = x_2^2 - 2x_1 - 1 \qquad z_2 = x_1^2 - 2x_2 + 1$$

Using geometry (not quadratic programming), what values of $\mathbf{w}$ (without w_0) and b specify the separating plane $\mathbf{w}^{\mathrm{T}}\mathbf{z} + b = 0$ that maximizes the margin in the $\mathcal{Z}$ space? The values of w_1, w_2, b are:

[a] $-1,\ 1,\ -0.5$

[b] $1,\ -1,\ -0.5$

[c] $1,\ 0,\ -0.5$

[d] $0,\ 1,\ -0.5$

[e] None of the above would work.

12. Consider the same training set of the previous problem, but instead of explicitly transforming the input space $\mathcal{X}$, apply the hard-margin SVM algorithm with the kernel

$$K(\mathbf{x}, \mathbf{x}') = (1 + \mathbf{x}^{\mathrm{T}}\mathbf{x}')^2$$

(which corresponds to a second-order polynomial transformation). Set up the expression for $\mathcal{L}(\alpha_1 ... \alpha_7)$ and solve for the optimal $\alpha_1, ..., \alpha_7$ (numerically, using a quadratic programming package). The number of support vectors you get is in what range?

[a] 0-1

[b] 2-3

[c] 4-5

[d] 6-7

[e] >7

• Radial Basis Functions

We experiment with the RBF model, both in regular form (Lloyd + pseudo-inverse) with K centers:

$$\text{sign}\left(\sum_{k=1}^{K} w_k \ \exp\left(-\gamma \ ||\mathbf{x} - \mu_k||^2\right) + \ b \right)$$

(notice that there is a bias term), and in kernel form (using the RBF kernel in hard-margin SVM):

$$\text{sign}\left(\sum_{\alpha_n > 0} \alpha_n y_n \ \exp\left(-\gamma \ ||\mathbf{x} - \mathbf{x}_n||^2\right) + \ b \right).$$

The input space is $\mathcal{X} = [-1, 1] \times [-1, 1]$ with uniform probability distribution, and the target is

$$f(\mathbf{x}) = \text{sign}(x_2 - x_1 + 0.25 \sin(\pi x_1))$$

which is slightly nonlinear in the $\mathcal{X}$ space. In each run, generate 100 training points at random using this target, and apply both forms of RBF to these training points. Here are some guidelines:

- Repeat the experiment for as many runs as needed to get the answer to be stable (statistically away from flipping to the closest competing answer).
- In case a data set is not separable in the '$\mathcal{Z}$ space' by the RBF kernel using hard-margin SVM, discard the run but keep track of how often this happens, if ever.

- When you use Lloyd's algorithm, initialize the centers to random points in $\mathcal{X}$ and iterate until there is no change from iteration to iteration. If a cluster becomes empty, discard the run and repeat.

13. For $\gamma = 1.5$, how often do you get a data set that is not separable by the RBF kernel (using hard-margin SVM)? *Hint: Run the hard-margin SVM, then check if the solution has* $E_{\text{in}} \neq 0$.

[**a**] $\leq 5\%$ of the time

[**b**] $> 5\%$ but $\leq 10\%$ of the time

[**c**] $> 10\%$ but $\leq 20\%$ of the time

[**d**] $> 20\%$ but $\leq 40\%$ of the time

[**e**] $> 40\%$ of the time

14. If we use $K = 9$ for regular RBF and take $\gamma = 1.5$, how often does the kernel form beat the regular form (excluding runs mentioned in Problem 13 and runs with empty clusters, if any) in terms of E_{out}?

[**a**] $\leq 15\%$ of the time

[**b**] $> 15\%$ but $\leq 30\%$ of the time

[**c**] $> 30\%$ but $\leq 50\%$ of the time

[**d**] $> 50\%$ but $\leq 75\%$ of the time

[**e**] $> 75\%$ of the time

15. If we use $K = 12$ for regular RBF and take $\gamma = 1.5$, how often does the kernel form beat the regular form (excluding runs mentioned in Problem 13 and runs with empty clusters, if any) in terms of E_{out}?

[**a**] $\leq 10\%$ of the time

[**b**] $> 10\%$ but $\leq 30\%$ of the time

[**c**] $> 30\%$ but $\leq 60\%$ of the time

[**d**] $> 60\%$ but $\leq 90\%$ of the time

[**e**] $> 90\%$ of the time

16. Now we focus on regular RBF only, with $\gamma = 1.5$. If we go from $K = 9$ clusters to $K = 12$ clusters (only 9 and 12), which of the following 5 cases happens most often in your runs (excluding runs with empty clusters, if any)? Up or down means strictly so.

[**a**] E_{in} goes down, but E_{out} goes up.

[**b**] E_{in} goes up, but E_{out} goes down.

[**c**] Both E_{in} and E_{out} go up.

[**d**] Both E_{in} and E_{out} go down.

[**e**] E_{in} and E_{out} remain the same.

17. For regular RBF with $K = 9$, if we go from $\gamma = 1.5$ to $\gamma = 2$ (only 1.5 and 2), which of the following 5 cases happens most often in your runs (excluding runs with empty clusters, if any)? Up or down means strictly so.

[**a**] E_{in} goes down, but E_{out} goes up.

[**b**] E_{in} goes up, but E_{out} goes down.

[**c**] Both E_{in} and E_{out} go up.

[**d**] Both E_{in} and E_{out} go down.

[**e**] E_{in} and E_{out} remain the same.

18. What is the percentage of time that regular RBF achieves $E_{\text{in}} = 0$ with $K = 9$ and $\gamma = 1.5$ (excluding runs with empty clusters, if any)?

[**a**] $\leq 10\%$ of the time

[**b**] $> 10\%$ but $\leq 20\%$ of the time

[**c**] $> 20\%$ but $\leq 30\%$ of the time

[**d**] $> 30\%$ but $\leq 50\%$ of the time

[**e**] $> 50\%$ of the time

- **Aggregation**

19. Given two learned hypotheses g_1 and g_2, we construct the aggregate hypothesis g given by $g(\mathbf{x}) = \frac{1}{2}\left(g_1(\mathbf{x}) + g_2(\mathbf{x})\right)$ for all $\mathbf{x} \in \mathcal{X}$. If we use mean-squared error, which of the following statements is true?

[**a**] $E_{\text{out}}(g)$ cannot be worse than $E_{\text{out}}(g_1)$.

[**b**] $E_{\text{out}}(g)$ cannot be worse than the smaller of $E_{\text{out}}(g_1)$ and $E_{\text{out}}(g_2)$.

[**c**] $E_{\text{out}}(g)$ cannot be worse than the average of $E_{\text{out}}(g_1)$ and $E_{\text{out}}(g_2)$.

[**d**] $E_{\text{out}}(g)$ has to be between $E_{\text{out}}(g_1)$ and $E_{\text{out}}(g_2)$ (including the end values of that interval).

[**e**] None of the above

- **Hints**

20. Let the target function $f : R^d \to R$ be the credit line based on the customer data $\mathbf{x}$, and assume $f(\mathbf{x})$ is monotonically non-decreasing in the annual salary, one of the variables in the input $\mathbf{x}$. If the transformation of $\mathbf{x}$ that increases the salary variable and keeps the rest of the variables in $\mathbf{x}$ unchanged leads to a transformed input $\mathbf{x}'$, which of the following error measures applied to a virtual (unlabeled) example $(\mathbf{x}, \mathbf{x}')$ would capture enforcing the hint on that example?

[**a**] $g(\mathbf{x}) - g(\mathbf{x}')$

[**b**] $g(\mathbf{x}') - g(\mathbf{x})$

[**c**] $(\ g(\mathbf{x}) - g(\mathbf{x}')\ )^2$

[**d**] $\text{ReLU}(\ g(\mathbf{x}) - g(\mathbf{x}')\ )$

[**e**] $\text{ReLU}(\ g(\mathbf{x}') - g(\mathbf{x})\ )$